

Job Access and Spatial Equity of a Toll Road

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Abstract: This chapter describes an evaluation of the job access and spatial equity impacts of the Cipularang toll road in the Jakarta – Bandung corridor in Indonesia, which was constructed in 2005. An ArcGIS, transport demand model extension, is used to obtain travel time and generalised cost estimates. The potential access measure and Shen access index are used to measure access with and without job competition. Spatial equity is examined based on the access measures using the Gini coefficient, Palma ratio, and a two-step cluster analysis. The analysis shows that the construction of the Cipularang toll road has reduced travel time in the whole region by 13% and potential job access increased by 5%. However, the toll road also increases job competition as more workers can access jobs available in the job-poor area in between Jakarta and Bandung, resulting in a small decrease in the number of accessible jobs per worker. This chapter concludes that the construction of the toll road has no impact on spatial equity as measured by the Gini coefficient and the Palma ratio.

Introduction

Access metrics estimating the number of activities that can be reached within a certain area are often used in spatial equity analysis as they are able to capture the quality of transport infrastructure and the spatial distribution of the activities (land use).¹ However, applications of access metrics in ex-post evaluations of road investments are rare, in particular for developing countries.

Keywords: Access; Equity; Employment; Job competition; Toll road

¹ (Geurs and Van Wee 2004).

A new road can have diverse equity impacts, ranging from access to social opportunities to regional economic developments. A particular example is the construction of toll roads, which allow users to have a typical choice between a quicker but expensive route and a slower but inexpensive route. The additional costs and changes in travel time can be unevenly distributed across a population, according to socioeconomic level. There can even be groups for which the new road has detrimental effects. For example, high-income commuters in Cape Town, South Africa are more likely to benefit from toll roads whereas lower-income commuters choose alternative non-tolled routes.²

² (van Dijk et al. 2015).

This chapter presents an ex-post evaluation of the impact of a toll road on job access and spatial equity in a developing country context. We used an ArcGIS, transport demand model extension, to estimate the transport impacts of the introduction of the Cipularang toll road in the Jakarta – Bandung corridor in Indonesia.

The Cipularang toll road links two major Indonesian cities, Jakarta, and Bandung. In 2004, 78.3 million passengers passed through the Jakarta - Bandung corridor, making it one of the busiest passenger transport routes in Indonesia. The distance between the cities is about 180 km and the modal split is dominated by car traffic.³ Empirical evidence also suggests that the introduction of this toll road corridor induced positive responses from the real estate market.⁴

³ (Lubis et al. 2005).

⁴ (Dorodjatoen 2009).

In the Indonesian context, ex-post evaluation of transport infrastructure investments, as in many countries across the globe, has not yet received much attention. Ideally, transport policy should follow the decision-making cycle,⁵ in which, needs or rationales are identified first to define the objectives or the desired transport system or system level. Then, an ex-ante evaluation (appraisal) is conducted to identify possible transport policies or projects. Subsequent steps (monitoring, ex-post evaluation and feedback) after the policy has been implemented are often neglected. An ex-post analysis is important as investigating past successes and failures improves the quality of appraisal for upcoming projects and enhances accountability. Many countries require ex-post evaluation of transport projects, however, few actually enforce this requirement, and this failure is often related to a lack of dedicated funding for such evaluations and limited availability of relevant data.⁶

⁵ (Emberger et al. 2008).

⁶ (Worsley 2017).

Questions

This chapter addresses the following question:

- What is the impact of the toll road on job access and spatial equity?

Given the importance of toll roads in the development of the major road network in Indonesia, this chapter focuses on the ex-post evaluation of the Cipularang toll road. This chapter conducts an empirical calculation of several access measures to distinguish between the effects on jobs and working population distribution. The Gini coefficient, Palma ratio, and a two-step cluster method are also applied to gain insight into the distribution of the impacts on job access across the region.

Methods

To assess the impact of the Cipularang toll road on access and spatial equity, we applied three stages of analysis in this chapter. Firstly, we created an ArcGIS transport network, then the 4-step transport model extension Traffic Analyst for ArcGIS⁷ was used and calibrated to produce traffic flows, travel time and generalised cost matrices in a scenario with and without the toll road. Next, we developed a 24-hour traffic simulation, which included the toll road. The use of the transport demand model enabled us to consider congestion effects by adding capacity restraints in the traffic model. We employed road network data from OSM⁸ as it provided road network data up to the local level. Applying four-step models is challenging because of data limitations. It is important to note as well that our model aimed to predict travel times in the Jakarta-Bandung region and, specifically, traffic flow on the corridor of the Cipularang toll road, thus we ignored traffic on other roads.

Secondly, we calculated access indicators by using the generated travel time and cost matrices, considering the number of jobs and the size of the working population. Thirdly, we conducted a spatial equity analysis based on the equity indices (Gini coefficient and Palma ratio) and spatial distribution of access among districts with a two-step cluster method. The unit of analysis in this chapter is districts in the study area.

⁷ (Rapidis 2018).

⁸ (OpenStreetMap contributors 2016).

Traffic simulation

The traffic simulation was conducted through the conventional 4-step model. Firstly, a number of trips were generated in each district by considering the population size and the proportion of residential area as production parameters and the number of jobs and the proportion of industrial and commercial areas as attraction parameters. The parameters were estimated from a regression analysis using population and land use data from the Jakarta metropolitan area.⁹

⁹ (JICA 2004).

In the second phase, the balanced-generated trips were distributed using the Furness method. This method also used a deterrence function, which was calculated from the gravity parameter and road traffic cost matrix. The gravity parameter assumes that a higher travel time will result in fewer trips made. We calculated it from the Indonesian labour force travel survey that mapped the mobility of labour forces in the study area in 2015.¹⁰

¹⁰ (BPS 2015).

Furthermore, road traffic cost was calculated from generalised cost considering the travel time between districts (using Google Maps API), fuel cost for each type of vehicle and value of time. The value of time was derived from a stated choice experiment survey conducted in the Jakarta-Bandung region for 1,600 respondents.¹¹ Next, a growth factor forecasting calculation (Furness method) was performed on the balanced generated trips and converted into an origin-destination (OD) matrix.

¹¹ See (Andani et al. 2021) for more details about this experiment.

In the third phase, OD matrices for three modes were estimated (car, heavy vehicle, and motorcycle), using the proportion of mode share in the area. Lastly, the road traffic assignment was carried out, assigning the generated OD matrix to the road network. A user equilibrium algorithm was used in this assignment. This last step produced the best travel route, the amount of traffic and the costs (travel time, distance, and generalised cost) for the pair of zones.

Access measurements

Access, is defined as the potential of opportunities for interaction.¹² There are many definitions afterwards. On the basis of a large number of studies,¹³ were able to distinguish four main elements to define access: transport, land use, temporal, and individual components. Access measures can be also categorised into: (1) infrastructure-based access, (2) location-based access, (3) person-based access, and (4) utility-based.¹⁴

¹² (Hansen 1959).

¹³ (Geurs and Van Wee 2004).

¹⁴ See more details in (Geurs and Van Wee 2004).

In this chapter, we applied infrastructure-based measures to assess the performance of the road network as well as location-based measures. The infrastructure-based measure focuses on the performance or service level of transport infrastructures, such as the length of infrastructure networks, level of congestion and average travel speed on the road network. It is worth noting that estimating travel time without considering capacity restraints and traffic flow will result in underestimation. Hence, several researchers also developed a traffic model to evaluate road infrastructure investment.¹⁵

¹⁵ For example (van Dijk et al. 2015, Xiong et al. 2015).

Furthermore, we estimated two access measures. Firstly, the standard job potential access measure was calculated by weighting opportunities in a certain area by using an attribute of attraction (population, facilities, etc.) and a measure of impedance (distance, travel time, cost). Secondly, the Shen access index was estimated to incorporate the spatial distribution of jobs and the working population competing for these jobs.¹⁶ Thus, we evaluated the combined effects of transport elements and job distribution, augmented by the inclusion of congestion and competition effects.

¹⁶ (Shen 1998).

Average travel time and generalised cost Average travel time and generalised cost $\bar{c}_r$ were calculated by summing up the travel time or generalised costs for all the trips from one origin (i) in situations with the Cipularang toll road ($r = 1$) and without it ($r = 0$), and then dividing this by the total amount of trips possible (n) from that origin.

$$\bar{c}_r = \frac{\sum_i c_{i,r}}{n_{c_{i,r}}} \quad (13.1)$$

The monetary cost in the link was determined from the generalised cost, taking into account fuel cost and value of time (V_t).

The generalised cost in link l ($C_{g,l}$) was calculated using the following formula:

$$C_{g,l} = C_{f,l} \cdot d_l + C_{\tau,l} \cdot d_l + C_{t,l} \cdot V_t \quad (13.2)$$

where $C_{f,l}$ is fuel cost (IDR per kilometre), d_l is the length of the link (in kilometre), $C_{\tau,l}$ is toll cost (IDR per kilometre), $C_{t,l}$ is travel time (hour), and V_t is the value of time (IDR per hour). A list of all variables can be seen in Table 13.3.

Table 13.1: Goodness of fit (R^2) of fitted distribution functions

Impedance	Distribution function			
	Log-logistic	Gaussian	Negative exponential	Power
Time	1.00	0.99	0.99	0.86
Generalised cost	1.00	1.00	0.97	0.77

Potential job access Job access denotes the ease that spatially distributed jobs can be reached from a given location. The most used potential access measure is based on Hansen's access formula,¹⁷ as follows:

¹⁷ (Hansen 1959).

$$S_{i,r} = \sum_j E_j \cdot f(c_{ij,r}) \quad (13.3)$$

Here, S_i is the access for location i with toll road ($r=1$) or without ($r=0$), E_j is the number of relevant jobs in location j , c_{ijr} is the cost (i.e. travel time or generalised cost) of a trip from i to j with toll road ($r = 1$) or without ($r = 0$), and $f(c_{ijr})$ is the cost decay function measuring the spatial separation between locations i and j .

The cost decay function reflects the friction in connecting zones i and j in the network and is negatively correlated with the attractiveness of the destination (i.e. jobs). The access results, therefore, are profoundly affected by the decay function. Various types of cost decay were tested for the access analysis such as negative power, negative exponential, Log-logistic and the Gaussian specification (see the comparison in Table 13.1 and Figure 13.1). Note that the power function is not depicted in Figure 13.1 as the probability value is very low. Regarding the impedance parameter based on the household survey data, we found that the log-logistic cost decay function (Equation 13.4) produced the best fit with the travel data.

The log-logistic cost decay function is specified as:

$$f(c_{ij}) = [1 + \exp(a + b \cdot \ln(c_{ij}))]^{-1} \quad (13.4)$$

where a and b are parameters to be estimated. We estimated the parameters of separate log-logistic functions for both time and generalised cost (Table 13.2).

Potential job access with impedance function and competition (Shen index)

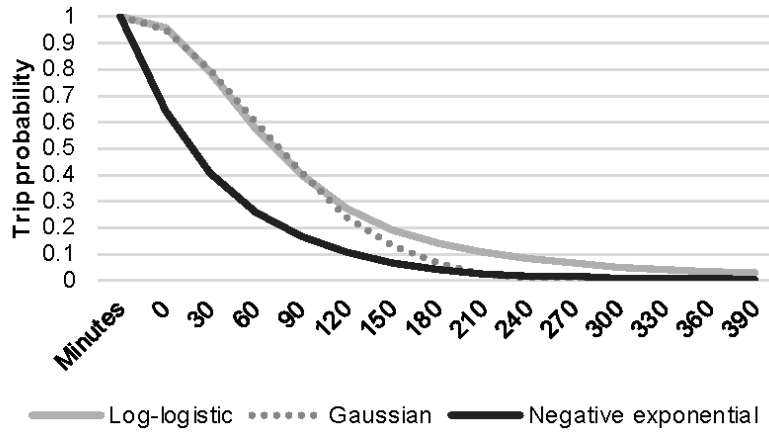


Figure 13.1: Net auto vs. net transit access to all jobs via auto within 30 minutes

To improve these access measures, we considered competition in the access measurement by including the potential demand for jobs. Potential job access measures the number of jobs within reach of a resident from an origin within a certain travel distance (potential supply) divided by the potential number of residents within reach of the same destination (potential demand).¹⁸ The Shen index is similar to the two-step floating catching area (2SFCA) method used in geography¹⁹ to access spatial access to health care. Several authors enhanced the 2SFCA method by for example incorporating distance decay.²⁰ In this chapter, we refer to this access measure as the ‘Shen index.’ If the Shen index is 1, then the number of jobs is proportional to the size of the working population. It can be calculated with the following formula:

$$A_{ir} = \sum_j \frac{f(c_{ij,r})}{D_j}, D_{jr} = \sum_k L_k f(c_{kj,r}) \quad (13.5)$$

Here, A_i is the Shen Index or the ratio between the potential job access of people living in location i , with toll road ($r=1$) or without ($r=0$) and the number of people seeking job opportunities in location j (D_j), E_j is the number of relevant jobs in location j , L_k is the number of the working population in the study area, and $f(c_{kjr})$ is the impedance function measure the spatial separation between k and j , with toll road ($r = 1$) or without ($r = 0$).

This chapter used data collected from the Indonesian database for policy and economic research²¹ to calculate job access. We used

Table 13.2: Parameters for log-logistic cost-decay function

Impedance	a	b
Travel time (C_t)	-11.55	2.50
Generalised cost (C_g)	-11.71	2.10

¹⁸ (Shen 1998).

¹⁹ (Radke and Mu 2000).

²⁰ e.g., (Luo and Wang 2003).

²¹ (WorldBank 2018).

Table 13.3: List of variables

Notation	Description
c	Cost (i.e. Travel time or generalised cost) of a trip
C_g	Generalised cost (IDR)
C_f	Fuel cost (IDR per kilometer)
d	The length of the link (in kilometer)
C_τ	Toll cost (IDR per kilometer)
C_t	Travel time (hour)
V_t	Value of time (IDR per hour)
S	Potential access of a location
E	Number of relevant jobs in a certain location
A	Shen Index (the ratio between the potential jobs access and the number of people seeking job opportunities)
D	People seeking job opportunities
L_k	Working population in the study area
l	Link in the network
r	Existence of a toll road [1,0]
i	Origin of a trip
j	Destination of a trip

employment data as a proxy for the number of jobs in the study area and defined employed people as persons who worked for pay or assisted others in obtaining pay or profit for at least one hour during the survey week. In addition, we used labour force data to calculate the number of job seekers in the study area. In this context, ‘labour force’ is defined as persons 15 years old and older who are working, are temporarily absent from work but had jobs, or who do not have work and are looking for work.²² This way, we exclude people of working age who are in schools, doing housekeeping, or not doing or looking for any profitable activity. This data is only available at the municipality level; to be able to conduct analyses at the district level, this data is extrapolated by using the population ratio. Note that due to data limitations, we were unable to distinguish between employed workers and job seekers and account for job types (e.g., by occupational class).²³

²² (BPS 2015).

²³ See (Smith et al. 2020).

Measuring spatial equity

access is normally used for equity analysis because it emphasises people and their interactions with places, and also captures both transport infrastructure and the spatial structure of the destinations (land use).²⁴ In the perspective of access, equity is commonly differentiated into spatial and social equity.²⁵ Spatial equity refers

²⁴ (Geurs and Van Wee 2004).

²⁵ (Van Wee and Geurs 2011).

to the geographical location of an individual, population group, or region, which is affected by an infrastructure project. Social equity is concerned with the impact of the infrastructure project on the economic or social condition of the individual, population group, or region. Since this chapter focuses on the geographical impacts of a toll road, spatial equity (i.e. its overall distribution, inequity at the extreme groups and whether job access is distributed evenly across the region) is addressed. To that end, we measured the Gini coefficient, Palma ratio, and the spatial distribution of access with a two-step cluster method.

Several statistical indicators can be used to assess the spatial equity implications of access impacts of road investments. A comprehensive study comparing various equity measures to assess the impact of road pricing policy concluded that using a single measure can lead to a biased evaluation.²⁶ The Gini coefficient,²⁷ which has been widely used to assess income inequity, is insensitive to any changes in measurement (scale-independent), which gives it a major advantage over other measures. The Gini coefficient also has been adopted to evaluate distributional impacts of access in some studies.²⁸

The Gini coefficient is commonly used to measure the distribution of income. It is a value between 0 and 1, where 0 is perfect equity and 1 is perfect inequity. We calculated the Gini coefficient to compare access levels with and without the toll road. We estimated the coefficient by using the trapezoidal Lorenz curve approximation, depicting the (ranked) cumulative share of access against the cumulative share of the corresponding population by district across the Jakarta – Bandung region. A Gini coefficient of 0.5 says that there is high inequality, but says nothing about the distribution of that inequality.²⁹ Since income per capita data is not available, monthly expenditure per capita data is used in the analysis. The spatial distribution of monthly expenditure per capita across districts in the study area can be seen in [Figure 13.3\(left\)](#).

Complementary to the Gini index, we also explored the Palma ratio.³⁰ Originally, the Palma ratio depicts the ratio of income shares between the richest 10% and the poorest 40%. Palma observed that changes in income are almost exclusively due to the changes in the share of the richest 10% and poorest 40% because the middle-income group between the richest and the poorest always capture approximately 50% of gross national income. He pointed

²⁶ (Ramjerdi 2006).

²⁷ (Gini 1936).

²⁸ (Lucas et al. 2016, Pritchard et al. 2019, Van Wee and Geurs 2011).

²⁹ (Banister 2018).

³⁰ (Palma 2011).

out that the stability of income share of the middle is a consistent finding across different data sets, countries and time periods.

³¹ Adopting methods from (Guzman and Oviedo 2018) and (Pritchard et al. 2019).

³² (Schwarz 1978).

In this chapter, we calculate the Palma ratio to assess the inequity between the average access of the richest 10% districts and the poorest 40% districts.³¹ A Palma ratio of 2.0 indicates that the top 10% (in terms of income) has twice the (job) access level as the lowest 40%. To account for the spatial distribution of job access, we clustered job access measures among districts in the study area by using a two-step cluster method in which the number of clusters was determined based on the minimum value of the Bayesian information criterion – BIC.³² Cluster analysis is commonly used in transport and land use domain studies to create groups of areas with homogeneous access measures and land use characteristics. This method enabled us to identify the characteristics of districts that are likely to benefit from the new toll road and classify them based on the access impacts.

³³ (Chi 2012).

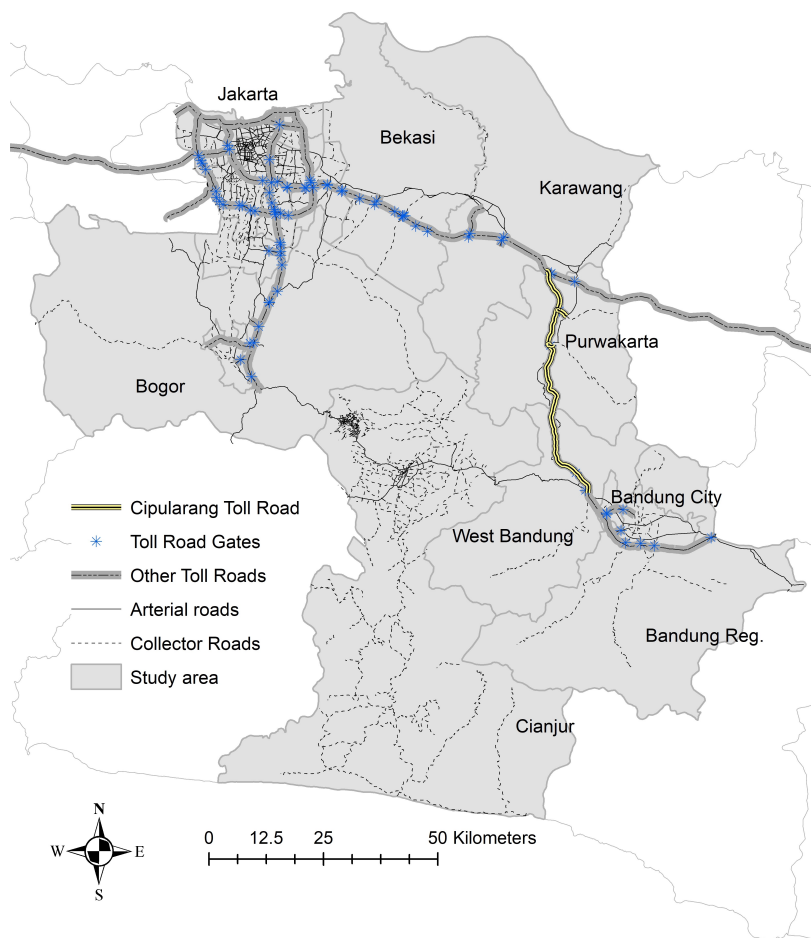
³⁴ (Di Ciommo and Lucas 2014, van Dijk et al. 2015).

Several variables are also included to identify the characteristics of districts that are likely to benefit from the new toll road and classify them based on the access impacts. According to Chi,³³ the impact of a new highway differ across rural, suburban, and urban areas. A new toll implementation would affect each income population segment differently.³⁴ Therefore, in the cluster analysis, we considered variables of residential land use growth and monthly expenditure per capita (as a proxy to income) to classify the districts. Furthermore, as the focus of this chapter, distribution of jobs and working population are also included in the cluster analysis.

Study area

The study area covers approximately 15,250 km² in the corridor between Jakarta and Bandung as shown in Figure 13.2. The name ‘Cipularang’ is a contraction of ‘Cikampek, Purwakarta and Padalarang.’ This toll road connects Jakarta and Bandung by connecting the Jakarta-Cikampek toll road and the Padalarang-Cileunyi (Padaleunyi) toll road (of which the Pasteur toll road in Bandung is also a part). The Cipularang toll road has five gates; three gates are located in the West Bandung Regency, and the other two are in the Purwakarta Regency. The study area covers of 16 cities or regency and 229 districts.

Figure 13.2: Study area



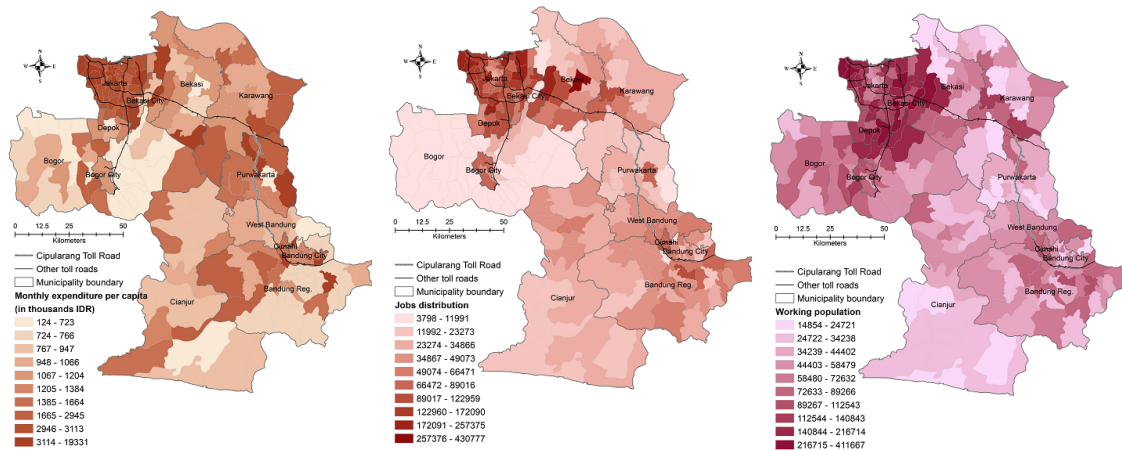


Figure 13.3: Spatial distribution of (left) monthly expenditure per capita, (center) jobs, and (right) working population

Between 2004 and 2013, residential land use increased by 2.5% and industrial land use by 1%.³⁵ The emergence of industrial and residential areas contributed to the increasing access, although jobs and the working population are still concentrated in the Jakarta and Bandung metropolitan area, as shown in Figure 13.3.

³⁵ Andani et al. 2019.

Findings

This section presents the results of the traffic simulation (average travel time and generalised cost), for potential job access and the spatial distribution of these values across the region.

Travel time estimation

Several performance measures were applied to evaluate the accuracy of the match between the observed travel time, derived from the Google Maps API, and simulated travel time. The value of the RMSPE is 9.7%, and the value of the correlation coefficient is 0.90. This indicates that the error is small enough and the accuracy of the model high enough to be able to predict travel time between origin-destination pairs. As Figure 13.4 shows, the model results show a similar pattern to the observed travel times. The estimated traffic flows on the Cipularang toll road are in line with the observed traffic data, with a percentage error of 3.7%. Figure 13.5 shows that without the Cipularang toll road, there is less traffic in

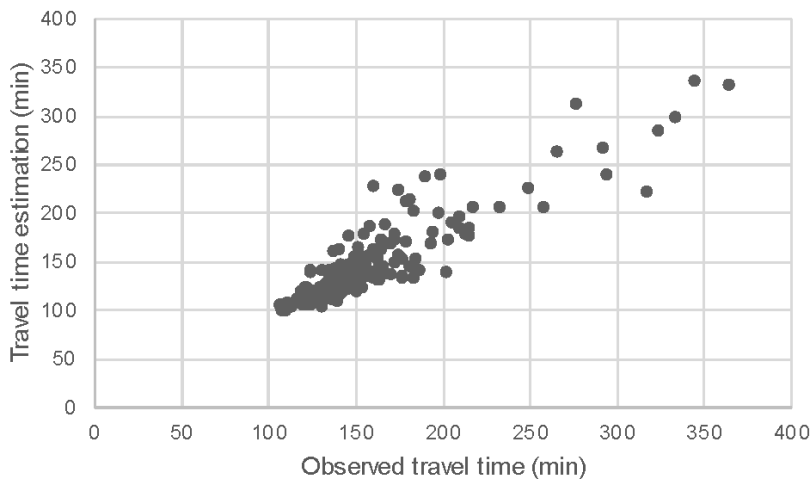


Figure 13.4: Comparison between observed and model results

Bandung and the northern part of this region. On the other hand, there is a significant increase in traffic on the other regional non-tolled roads.

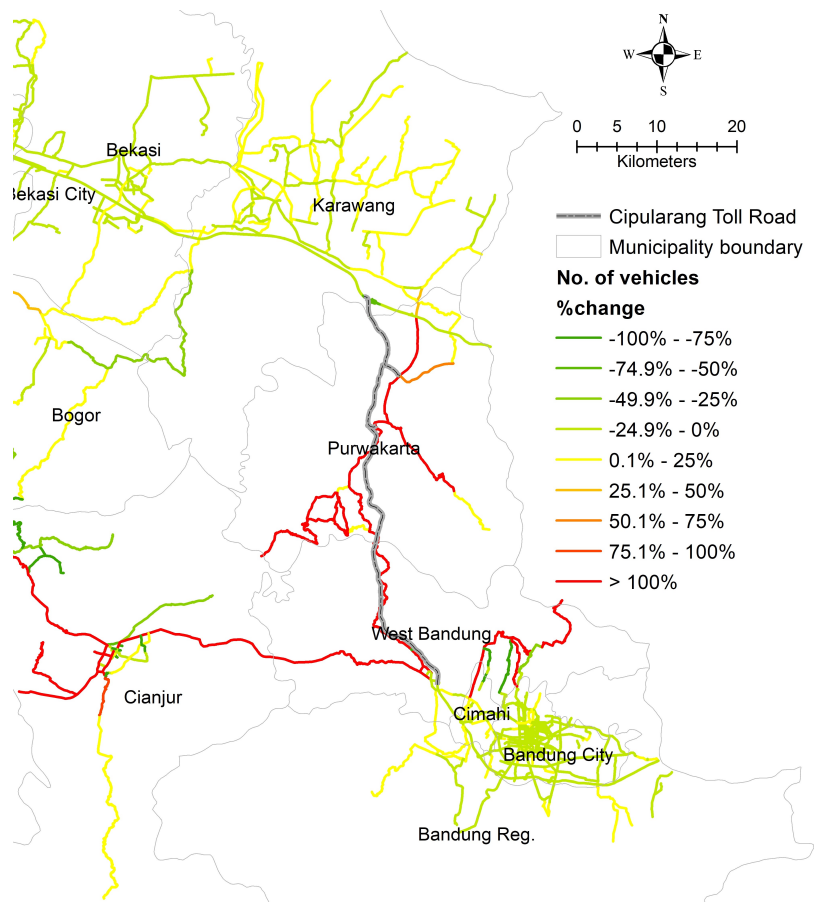
The average travel time between districts was estimated to be 154 minutes, similar to the observed travel time obtained from Google Maps API (155 minutes). The most strongly impacted regions are in the corridor including the north-western area in Jakarta and the south-eastern region in Bandung (see Figure 13.6).

Regions in the west, such as Bogor and Cianjur, appear to be least affected by the development of the Cipularang toll road. Overall, weighted by the traffic flow, the Cipularang toll road has reduced the average travel time in the whole region by 13%; see Table 13.4. Several districts in Purwakarta experienced a weighted average travel time reduction up to 25%. This can be explained by the fact that this area contains two toll gates. It is also situated between two employment centres in the area: Jakarta and Bandung. This finding supports the observation that the impact of toll mostly occurs in the area alongside the toll road.³⁶

³⁶ (Van Wee and Geurs 2011).

When we looked at specific origin-destination pairs, we found that the greatest reduction in travel time occurred between Cipendeuy (a district in the Bandung regency) and Teluk Jambe (a district in Karawang), which had almost 1.5 times greater travel times without the Cipularang toll road. The difference between the estimated average generalised cost with and without the toll road is not as great as the difference for the average travel time, given that

Figure 13.5: Changes in traffic flow without the Cipularang toll road



to use the toll road, users need to pay a fee, which increases the total travel cost. As observed from Figure 13.6, the estimated average travel time and generalised cost show a contrast between areas in the toll road corridor and areas at great distances from it.

Potential access measures

Table 13.5 presents the results of various access measures, weighted by the working population in each zone. access measures with time decay showed more significant changes than measures with generalised cost impedance. This indicates an overestimation of the impact as it is only taken into account the travel time. It is worth noting that the result of each access measures in this chapter is different from each other, though complementary to each other.

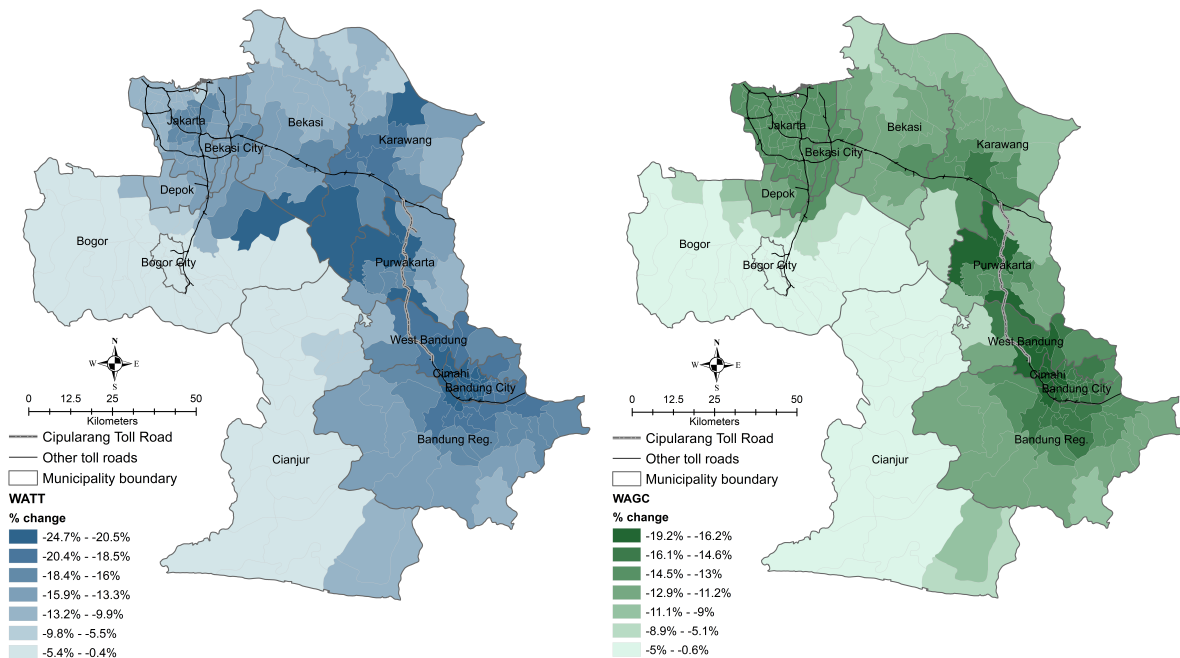


Figure 13.6: Relative difference of weighted average (left) travel time and (right) generalised cost between districts

Moreover, the potential access measures particularly highlight differences in job access between large urban agglomerations and rural areas.

Furthermore, the Shen index (Figure 13.7a) shows the proportion between the potential number of jobs that can be accessed and the potential demand for jobs. Several districts in Jakarta and Purwakarta have Shen indices equal to one, which indicates that the number of jobs and the size of the working population are in balance. A district in Bekasi, known as the largest industrial area in Indonesia, has a Shen index of more than 2.0, which signifies that there are more jobs available than workers. However, in contrast with other measures, the Shen index is lower in the scenario with the Cipularang toll road (see Figure 13.7a). The reason is that the Cipularang toll road not only increased job access, but also working population access. The Shen index decreases the strongest in the job-poor area between Jakarta and Bandung. The construction of the Cipularang toll road also enabled the working population in the two large cities to access new jobs emerging in the new industrial areas along the toll road.

Previous findings that the number of large and medium-sized industries has increased by almost 15% in Purwakarta and that the

Table 13.4: Descriptive statistics of estimated average travel time and generalised cost

Statistics	Travel Time (minutes)	Generalised Cost (1,000 IDR)
With Cipularang toll road		
mean	154.5	231.7
max	332.6	453.8
min	109.4	180.3
st. dev.	36.5	43.6
Without Cipularang toll road		
mean	167.5	233.4
max	335.1	454.3
min	129	180.6
st. dev.	35.5	43.6

³⁷ (Dorodjatoen 2009).

industrial relocation was due to the proximity to Jakarta and the availability of cheaper labour strengthen this result.³⁷ The growth of new settlement areas was also observed in Purwakarta. Dorodjatoen found that people from Bandung and Jakarta were the predominant residents of the new settlements. The construction of the Cipularang toll road enabled them to obtain less expensive housing in Purwakarta and commute daily to one of the two big cities.

Spatial equity evaluation

Equity indices Table 13.6 shows the results for both equity indices with the Shen index and with potential job access using generalised cost decay; both are weighted by the working population. access measures with time decay were not included in the equity analysis as it only took the travel time into account. This led to an overestimation

Weighted access measure	Scenario	Impedance	
		Time	Generalised cost
Working population access	w/ toll road	8,135,674	11,304,273
	Without toll road	7,804,450	11,264,201
	Relative change	4.2%	0.4%
Shen Index	w/ toll road	0.73	0.73
	Without toll road	0.78	0.74
	Relative change	-5.6%	-0.41%
Potential job access	w/ toll road	6,016,274	8,251,967
	Without toll road	5,755,050	8,221,007

Table 13.5: Results of access measures

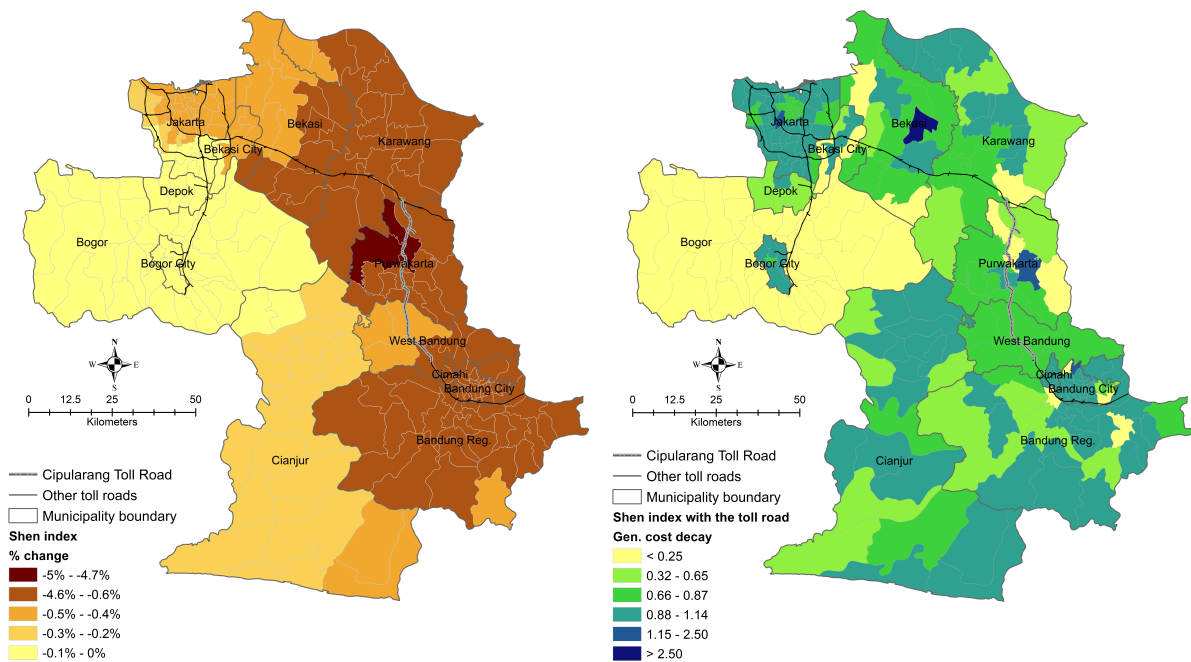


Figure 13.7: Relative changes in Shen Index with and without the toll road (left) and Shen Index with the toll road (right)

of the access impact of the toll road and, therefore, was not a good approach to measure the inequity.

In general, we found that the construction of the toll road had no impact on the average level of spatial equity as shown by the values of the Palma ratio and Gini coefficient with regard to potential job access, and of the Palma Ratio using the Shen index. However, we observed a slight increment of the Gini coefficient using the Shen index. This indicates that, overall, the introduction of the toll road has benefited high-to-medium income workers more than low-income workers, but the Palma ratio shows that people in the top 10% of incomes did not benefit more from the toll road than those in the lowest 40%.

However, this result is not fully representative, as mode choice was not taken into account. In Indonesia, low-income workers rely on motorcycles,³⁸ but motorcycles are not allowed on most toll roads. In practice, this means that higher-income workers benefit more than lower-income workers.³⁹

Cluster analysis The Gini index and Palma ratio (see the previous section) do not explain the spatial distribution of the changes in equity and the characteristics of districts that are most likely to

³⁸ (Herwangi et al. 2015).

³⁹ See Andani et al. (2021) for a more elaborate discussion on how the travel behaviour of various population segments is affected by the Cipularang toll road.

Table 13.6: Equity indices weighted by working population, with generalised cost decay

Access measures.	Scenario	Palma ratio	Gini coefficient
Shen index	w/ toll road	1.10	0.30
	w/out toll road	1.10	0.29
Potential job access	w/ toll road	1.14	0.38
	w/out toll road	1.14	0.38

benefit from the new toll road. This section, therefore, presents the results of an analysis of the spatial distribution of access carried out with a two-step cluster method.

As all types of potential access (labour market, job and Shen index) are highly correlated (>0.9), only the Shen index with generalised cost decay is included in the cluster analysis. We consider this access measure the most comprehensive of the measures tested in this chapter since it incorporates the competition effect and is calculated using generalised cost decay. Furthermore, we also took into account residential land use growth, monthly expenditure, number of jobs and working population of the areas in the analysis.

Figure 13.8 and show the cluster map and the profiles of each cluster containing the mean of each variable (cluster centre), respectively. We obtained three final clusters (based on minimum BIC value), which we distinguished these clusters as (1) affected areas, (2) unaffected and more urbanised districts and (3) unaffected and less urbanised districts.

Cluster 1 accounts for 2.6% of the 229 districts in the area. This cluster has the lowest relative change in the Shen index (-2%), while the changes in the other clusters are almost zero, thus this cluster experienced increased inequity. This can also be seen in the mismatch between the number of jobs and the size of the working population. This cluster can also be characterized as highly accessible, as it is situated close to the toll gates and has the highest reduction of average travel time relative to the situation before the construction of the Cipularang toll road (see Figure 13.6), with the highest expenditure per capita as well. This result also in line with the result of the Shen index, as shown in Figure 13.7(left). It is consistent with the findings of several studies that the most affected areas by the construction of new motorways were areas along the roads and close to the exits.⁴⁰

⁴⁰ e.g., (Ghani et al. 2016, Ji et al. 2014).

Variables	Cluster Centre			Combined
	Cluster 1 Affected districts	Cluster 2 Unaffected and more urbanised	Cluster 3 Unaffected and less urbanised	
Shen-index relative change	-2.02%	-0.34%	-0.48%	-0.49%
Monthly expenditure/capita (1,000 IDR)	8626	2473	1342	1750
Residential land use growth (2004-2013)	5.0%	5.2%	2.9%	3.4%
Working population	104,303	155,695	54,766	75,456
Number of jobs	14,526	138,027	36,204	55,201

Table 13.7: Final cluster centre

The shares of the number of jobs and the size of the working population in the other two clusters are somewhat proportional. However, Cluster 2 (unaffected and more urbanised districts) can be characterized as more urbanised due to its large number of workers and jobs, as well as its high rate of residential land use growth. This cluster accounts for 19.2% of the total districts in the region. Furthermore, Cluster 3 (unaffected and less urbanized districts) contains most districts in the study area (78%). This cluster is characterised by the lowest number of jobs and the smallest working population, the lowest average monthly expenditure, and the lowest residential land use growth.

These results highlight that the construction of a new toll road can have significant spatial equity impacts on areas close to the toll gates. The decreased travel times and the emergence of new employment along the toll road corridor increases the number of workers who can access employment. This cluster result explains the results of both equity indices (Palma ratio and Gini coefficient).

Conclusions

We examined the impact of the Cipularang toll road in Indonesia on job access and spatial equity by isolating the effect of the toll road. We applied a macroscopic traffic simulation model to obtain valid estimates of travel time and generalised cost. Job competition, represented by the Shen index, was also measured to reflect the distributions of employment and working population. We then assessed the equity impact by using the Palma ratio, Gini coefficient and spatial distribution of the access changes with a two-step cluster analysis.

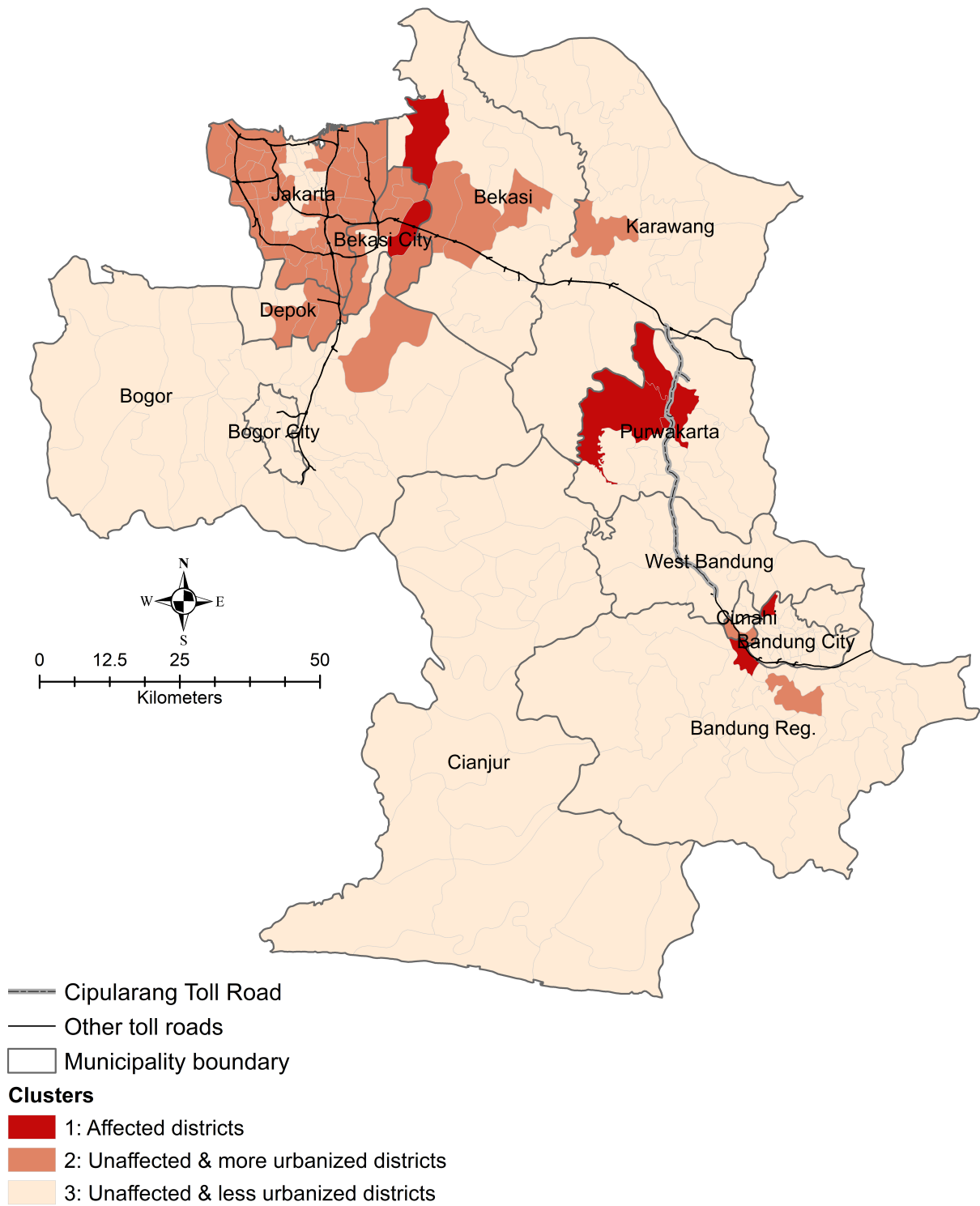


Figure 13.8: Cluster map

We found that area along the toll road and close to the toll gates are the most affected by the construction of the Cipularang toll road. Our analysis shows that the construction of the Cipularang toll road has reduced the average travel time in the Jakarta-Bandung area by 13%. For specific origin-destination pairs, the travel time is now up to 1.5 times shorter. The access analysis shows that the choice of indicator affects the results. Employment and working population access have increased by 5%. However, the toll road also increases job competition as more workers can access jobs available in the job-poor area in between Jakarta and Bandung, resulting in a small decrease in the number of accessible jobs per worker. Most affected areas have a relatively large working population compared to the number of jobs and experienced residential land use growth. We can conclude that a new toll road in developing countries with un-mature road networks can have positive as well as negative impacts on the areas close to it.

This chapter adds to the literature by examining job access and spatial equity impacts of a major toll road in a developing country context with the addition of congestion effects and stated choice-based value of time in the traffic simulation. The findings of this chapter highlight the potential value added of access analysis in the transport planning practice. access analysis can help to improve the integration of land use and transport planning with the goal of enabling all individuals, regardless of their social class or ability, to participate in desired activities in the area using the transport system.

Next

There remain opportunities to extend this research.

Firstly, we did not examine the impact of the toll road on spatial developments, e.g., distribution of employment, population or housing costs. The inclusion of these data in the measurement certainly will affect the potential job access measure.

Secondly, we were not able to estimate job access for different worker groups as reliable sector-level employment data or detailed data on the working population is not available at a high spatial resolution in Indonesia.

Finally, this chapter did not consider the complexity and differences in individual travel behaviour. It is important to incorporate the availability of transport modes at the individual level, as in Indonesia, low-income workers rely on motorcycles,

which are not allowed on most toll roads. Therefore, higher-income workers may benefit more than lower-income workers.